EOLLM Vision SFT

Teaching Qwen3.5-4B to Understand Cities

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The Task

Given: Satellite + street-level imagery of a location

Ask: Multiple-choice questions about urban properties

Answer: A single letter (A, B, C, or D)

Scale	Count
Cities	40 worldwide
Topics	14 urban categories
Train samples	26,931
Validation samples	6,984

Topics covered:

Land use, building height, road type, road surface, urban density, green space, junction type, amenity richness, transit density, camera direction, mismatch detection (4 variants)

Setup

Model

Base	Qwen3.5-4B
Method	bf16 LoRA
Trainable	38.8M / 4.58B **(0.85% of params)**

Hardware

GPU	RTX 5090 (32 GB)
Peak VRAM	17 GB (52%)
Wall clock	96 minutes

Why these choices?

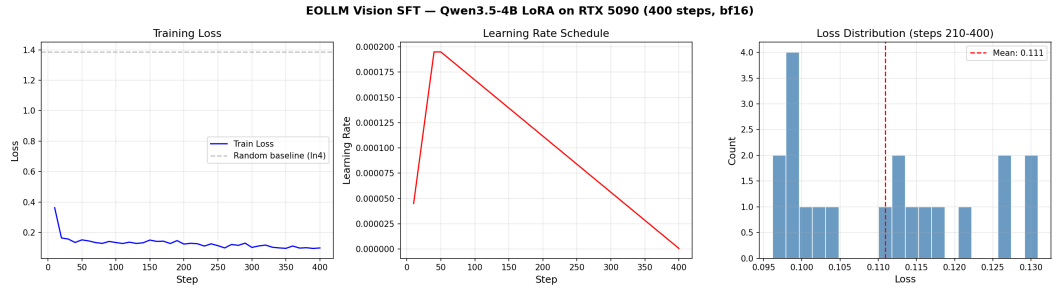
Choice	Reason
bf16, not 4-bit	Qwen3.5 degrades with quantization
LoRA r=16	Higher risks overfit on 27k samples
All-linear	Vision + language both need adaptation
LR 2e-4	Standard LoRA starting point

Hyperparameters

Parameter	Value	Rationale
Batch size	$4 \times 6 = 24$ effective	Largest before OOM
Learning rate	$2 \times 10^{-4} \rightarrow 0$	Linear decay with 40-step warmup
LoRA rank	16 (alpha = 16)	Balanced capacity vs overfit risk
max_seq_length	8192 tokens	p99 = 1,535; generous headroom
Image max edge	768 px	Profile-matched to RTX 5090
Optimizer	AdamW 8-bit	VRAM efficient
Training steps	400	$\sim$35% of 1 epoch

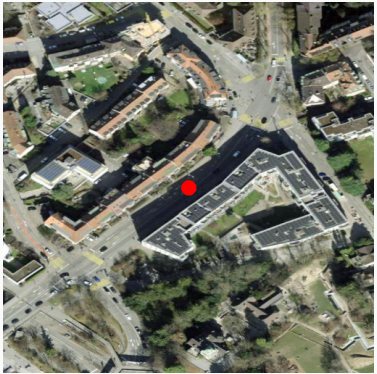
Token budget measured on 300 samples using the actual Qwen3.5 processor.
Batch size determined by forward+backward probe at bs = 1, 2, 4, 8.

Training Curves



Loss drops from **0.36** → **0.10** in 400 steps (random baseline ≈ 1.39). Only 35% of one epoch completed.

Example 1 — Land Use Classification (Zurich)



Satellite (marked)

Q: Based on the imagery, what best describes how this area is primarily used?

A. Residential ← **correct**
B. Industrial & Manufacturing
C. Transportation & Infrastructure
D. Mixed Use

Model	Answer	
Base Qwen3.5-4B	C	✗
Finetuned (400 steps)	**A**	✓

Example 2 — Satellite–Street View Matching (Buenos Aires)

Satellite (marked)

Q: Which of the four street view sets is the correct match?

A. Top-Left

B. Bottom-Right

C. Bottom-Left ← correct

D. Top-Right

Model	Answer	
Base	"To determine the correct match, we need to analyze..."	✗
Finetuned	**C**	✓

Base model generates explanations instead of answering.

Example 3 — Land Use (Mexico City)

Satellite (marked)

Q: How is this land primarily utilized?

A. Retail & Shopping

B. Mixed Use ← correct

C. Open Space & Recreation

D. Commercial & Office

Model	Answer	
Base Qwen3.5-4B	C	✗
Finetuned (400 steps)	**B**	✓

Results Summary

	Base Qwen3.5-4B	Finetuned (400 steps)
Eval accuracy (3 samples)	0/3 (0%)	**3/3** (100%)
Training loss	—	0.36 → **0.10**
Answers format	rambling text	single letter

What we trained:

- 0.85% of parameters (LoRA)
- 400 steps (∼35% of epoch 1)
- 96 minutes on a single GPU

What's left:

- Full 2-epoch training run
- Validation loss tracking
- Per-topic accuracy breakdown
- Benchmark evaluation (5,240 samples)